

Background

This report is about a real-world example of AI deployment (e.g., facial recognition) where I will evaluate the ethical implications using at least 5 AI ethics concepts from class.

Introduction

Facial recognition has become an increasingly common feature of everyday digital life. Instead of entering a password or personal identification number (PIN), users can simply look at their smartphones to unlock them. Now, examining the ethical concerns of facial recognition through the principles of beneficence, non-maleficence, autonomy, justice, and explicability demonstrates why effective AI governance must extend beyond technical performance and legal compliance.

1. **Beneficence**

The first ethical principle applicable to this case is *beneficence*, which emphasizes the responsibility to promote benefits and human well-being.

Facial authentication offers clear advantages. It can provide convenient access to a device and reduce dependence on passwords. However, beneficence should extend beyond convenience. A technology intended to benefit users should also protect their legitimate interests and support their ability to maintain meaningful control over their personal information.

2. **Non-maleficence**

A closely related ethical principle is *non-maleficence*, which emphasizes the obligation to avoid or minimize harm.

Facial recognition systems raise concerns about unauthorized access, cybersecurity breaches, unintended data linkage, and the potential misuse of biometric information.

3. **Autonomy**

A third important ethical principle is *autonomy*, which refers to an individual's ability to make informed and independent decisions.

In the context of facial recognition, autonomy requires that users have meaningful choices about whether their biometric information is collected and used.

4. **Justice**

The principle of *justice* is particularly significant when considering algorithmic bias and discrimination.

A facial recognition system may produce errors, but the ethical concern becomes more serious when errors or disadvantages are not distributed equitably across populations. In such circumstances, a technical limitation can become an issue involving discrimination, equality, and social justice.



5. **Explicability**
   The fifth principle is *explicability*, which concerns both understanding how AI systems operate and establishing responsibility for their outcomes.
   Effective governance should establish responsibilities before harm occurs. Organizations involved in developing, implementing, or managing AI technologies should have clearly defined responsibilities regarding privacy, security, monitoring, auditing, complaints, and remedies.

Conclusion

Facial recognition demonstrates why data science ethics cannot be separated from technological innovation.
The objective should not necessarily be to reject facial recognition or artificial intelligence. Instead, the goal should be to ensure that these technologies are developed and governed in ways that remain *human-centred and ethically responsible*.